

Euclid Data in the Cloud: Access, Analysis, and Science Opportunities

ENSCI (Euclid NASA Science Center at IPAC)

IRSA (NASA Infrared Science Archive)

248th AAS — June 14th, 2026



Today's Agenda

≈ 90 minutes — intro slides, then three notebooks in Fornax

1

Intro & framing

Workshop setup, Euclid + ENSCI, Q1 release, data-product acronyms, Fornax sign-up.

SLIDES

~ 15 min

2

Euclid data in the cloud

TAP / SIA / direct-S3 on a single EDF-N source — MER catalog, mosaic cutout, SIR 1D spectrum, SPE line features.

NOTEBOOK

~ 20 min

3

Galaxy clusters

Same toolkit on a high-richness Q1 cluster — ADQL JOIN across MER + PHZ, DBSCAN vs control field, over-density on H-band.

NOTEBOOK

~ 10 min

4

AGN hands-on

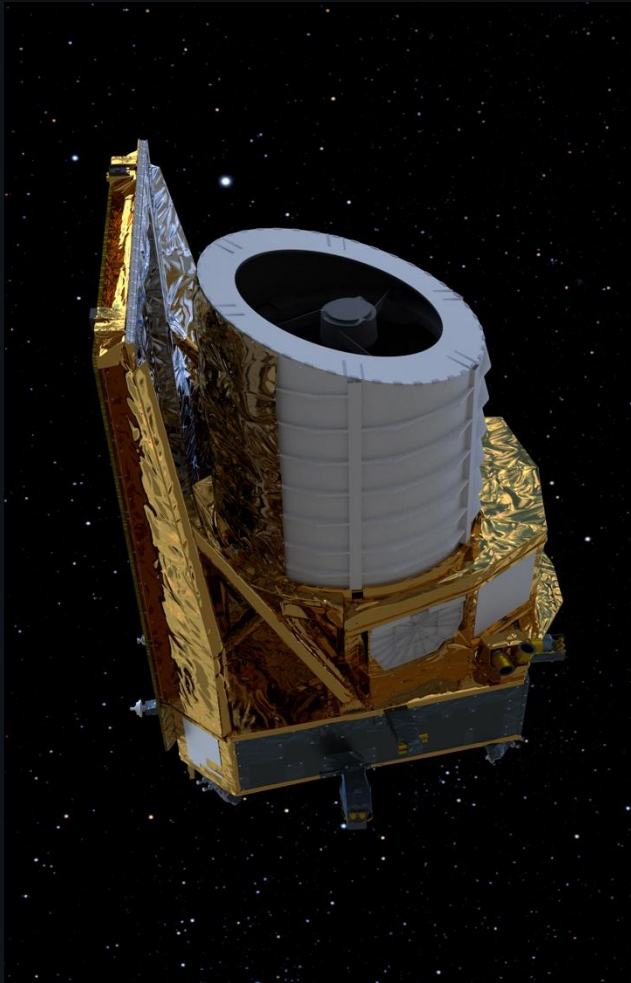
Project ~5k EDF-N AGN candidates onto a pre-trained color-SOM — compare selection methods, verify with three live cloud reads.

NOTEBOOK

~ 45 min

Euclid & ENSCI

ESA mission mapping the dark Universe — with NASA's US science center at IPAC



THE MISSION

Euclid

ESA M-class mission

Studies the geometry and nature of the dark Universe via weak gravitational lensing and galaxy clustering.

- **Launched** July 1, 2023 / 6-year mission at L2
- **1.2 m telescope** at optical + near-IR (0.5–2 μm)
- **14,000 sq deg** survey, a third of the sky
- **Consortium** of 2,600+ across Europe, US, Canada, Japan

THE US SCIENCE CENTER

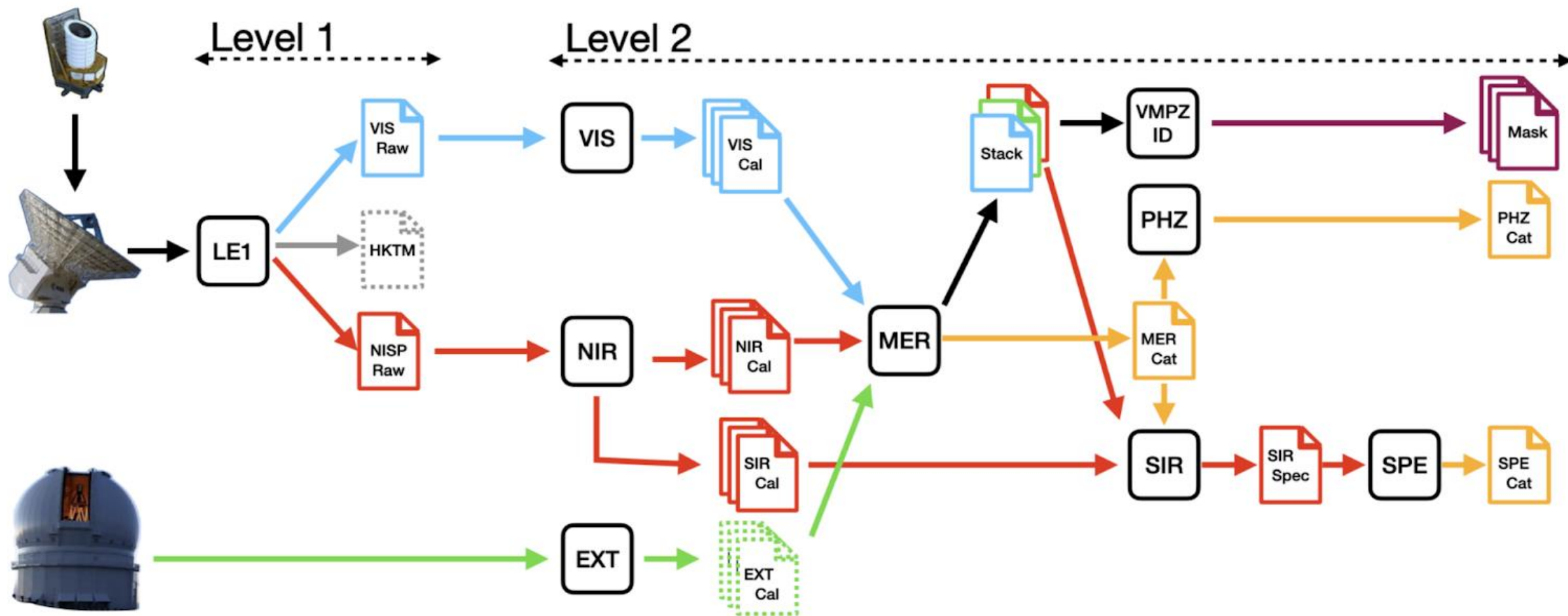
ENSCI

Euclid NASA Science Center at IPAC

Established by NASA at Caltech/IPAC in Pasadena to support US-based investigators using Euclid data.

- **Operates SDC-US** one of 11 data-processing nodes
- **US data archive** via NASA/IPAC IRSA
- **NASA contribution** 16 near-IR detectors + 3 nominated US science teams
- **User support** Help Center, documentation, tutorials

More at euclid.caltech.edu



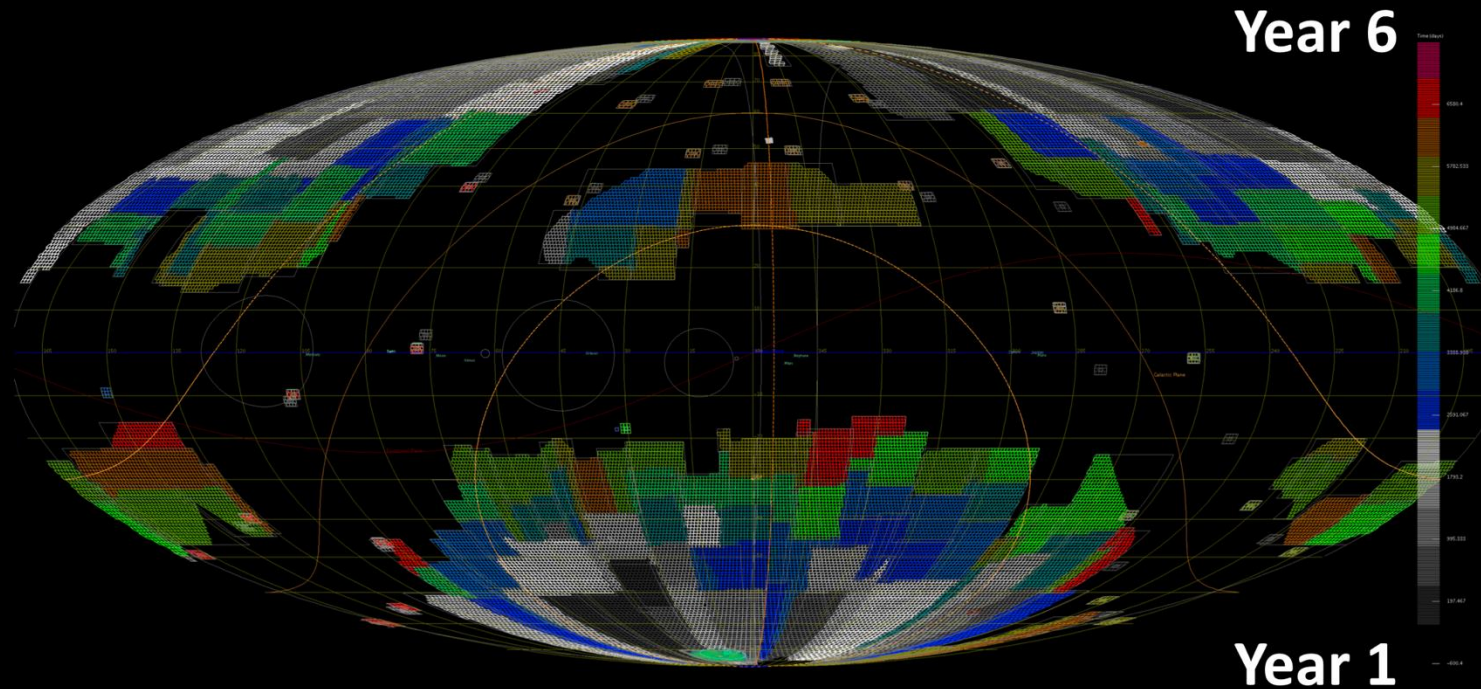
 : Processing Function

 : Data Product

 : Unpublished Data Product

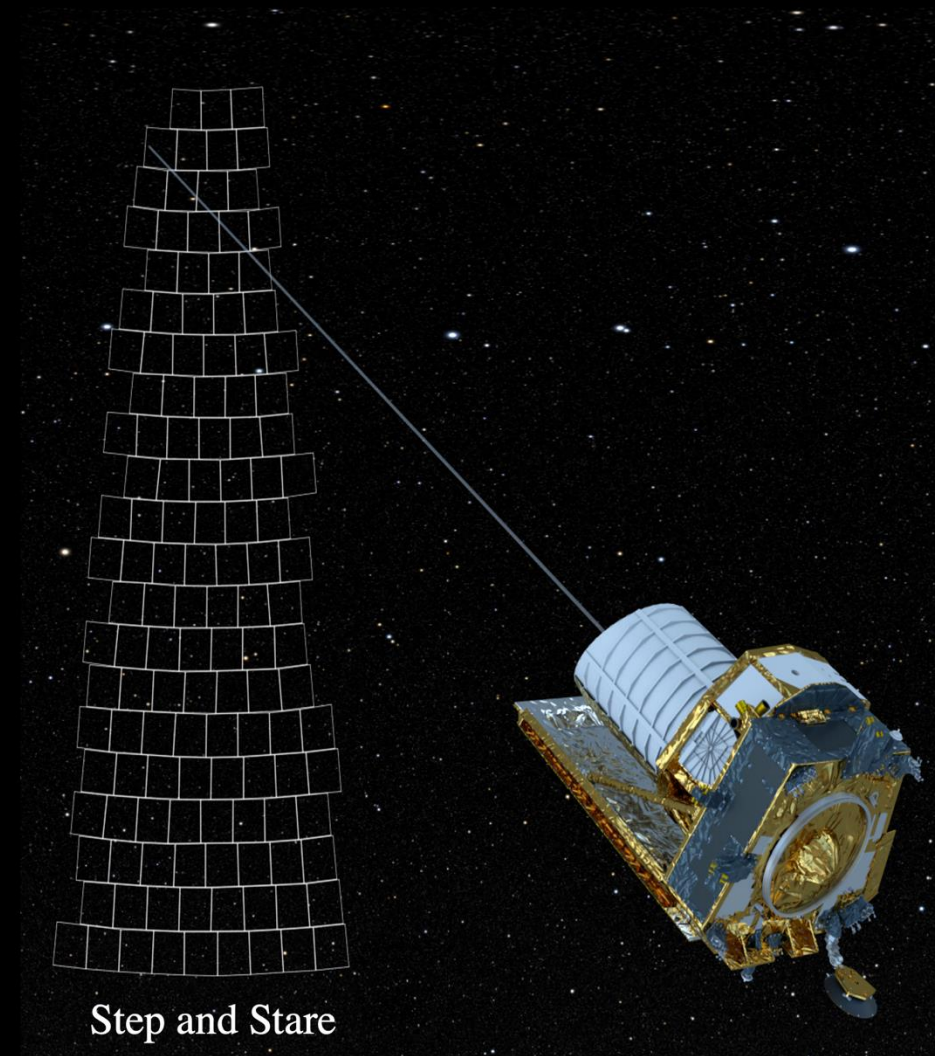
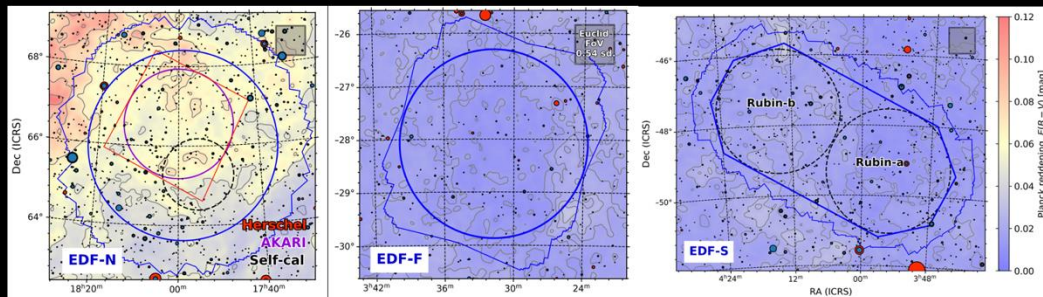
→ : VIS Data
 → : NISP Data
 → : EXT Data
 → : Multi λ Catalogs
 → : Multi λ Masks

Euclid Surveys



Year 1

$\sim 15,000 \text{ deg}^2$ WIDE survey + 3 deep fields $\sim 50 \text{ deg}^2$:



Data Release Schedule

Q1 — QUICK RELEASE 1

Released March 19, 2025

63 deg² over three Euclid Deep Fields at Wide Survey depth ~26 M detections.

- 23 deg² of **EDF South**
- 20 deg² of **EDF North**
- 10 deg² of **EDF Fornax**
- **Products:**
 - VIS + NISP imaging,
 - NISP spectroscopy,
 - catalogues & masks
- **Access:**
 - NASA/IPAC IRSA
 - ESA Euclid Science Archive

Mission Timeline

Year 1

Q1

2

DR1 ~ 2500 deg²

3

Q2

4

DR2 ~ 7500 deg²

5

Q3

6

Q4

DR3 ~ 15000 deg²

Useful Links

Euclid Q1 Data Product Description

euclid.esac.esa.int/dr/q1/dpdd

ESA Q1 Data Release Page

cosmos.esa.int/web/euclid/euclid-q1-data-release

Euclid Tutorial Notebooks

caltech-ipac.github.io/irsa-tutorials/euclid

Euclid Tutorial Videos

caltech-ipac.github.io/euclid-q1-webinar



NASA Fornax

JupyterLab in the cloud, next to NASA archive data

QUICK START

1. Open the quick-start guide

- docs.fornax.sciencecloud.nasa.gov/quick-start/
- or google “fornax quick start”

2. Make an account (approved instantly)

3. Log in to Fornax

4. Open a Jupyter Notebook

- Choose Medium • ~4 min to load

5. Click the “github extension”



6. Clone the workshop repo

- github.com/Caltech-IPAC/euclid-workshop
- Click “Download”



Scan to start

Quick-start guide

Q1 DATA RELEASE — MARCH 2025

63 deg² over three Euclid Deep Fields — VIS + NISP imaging, NISP spectroscopy, catalogues & masks.

FREE

BETA

BROWSER-BASED